

Design and Implementation of a PID-Based DC Motor Speed Control System Using MATLAB, Simulink, and Interactive GUI

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Design and Implementation of a PID-Based DC Motor Speed Control System Using MATLAB, Simulink, and Interactive GUI

Abstract

This project presents the modeling, control design, simulation, and graphical interface development of a DC motor speed control system using MATLAB and Simulink. A proportional–integral–derivative (PID) controller is implemented to regulate motor speed under varying load torque conditions. The system is validated through time-domain response analysis, disturbance testing, and real-time visualization using a custom-built MATLAB App Designer interface. The project demonstrates practical control system development from mathematical modeling to interactive user-level deployment.

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1. Introduction

In modern engineering, control systems are essential for regulating the behavior of dynamic systems across multiple industries, including robotics, industrial automation, automotive systems, and consumer electronics. Understanding system behavior is critical for ensuring stability, performance, and reliability. However, conventional simulations often rely solely on numerical plots, which can be difficult for stakeholders to interpret quickly.

This project demonstrates the **design, simulation, and real-time animated visualization of a control system** using MATLAB. By integrating real-time animation with control system simulation, the project provides an intuitive, visual understanding of system dynamics, controller performance, and transient behavior. This approach not only enhances comprehension but also creates a compelling presentation for potential clients or portfolio showcases.

2. Project Objectives

The primary objectives of this project are:

- Develop a **mathematical model** of a dynamic system (DC motor) suitable for control design and simulation.
- Design and implement **control strategies**, including PID and state-feedback controllers, to regulate system behavior. But for now only PID
- Simulate the system response under various operating conditions to analyze stability, transient response, and performance.
- Create a **real-time animated visualization** to demonstrate the system's dynamic response in an intuitive and interactive way.
- Prepare **professional documentation** including figures, animations, and explanations, suitable for client presentations or portfolio inclusion.

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3. Problem Statement

In practical systems, DC motor speed varies due to:

- Load torque disturbances
- Parameter uncertainties
- Voltage fluctuations

The objective is to design a control system that:

- Tracks a desired reference speed
- Minimizes overshoot
- Achieves fast settling time
- Maintains zero steady-state error
- Recovers quickly from disturbances

4. Control Strategy Overview

To achieve precise regulation of the DC motor speed, this project implements a **PID controller** (Proportional-Integral-Derivative), which is one of the most widely used control strategies in industry:

- **Proportional (P):** Reduces current error by applying a control proportional to the difference between desired and actual speed.
- **Integral (I):** Eliminates steady-state error by accumulating past errors.
- **Derivative (D):** Improves transient response by reacting to the rate of change of error.

The **open-loop response** will be shown first to illustrate the uncontrolled system, followed by the **PID-controlled response**, allowing comparison of performance.

5 Project Scope & Deliverables

5.1 Project Scope

- modeling of a DC motor system
- PID controller design and tuning

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- Numerical simulation of system response
- Mathematical Real-time animated visualization
- Documentation with figures, explanations, and animations

5.2 Deliverables:

- MATLAB source code: Modular and reusable simulation framework
- Figures and animations: Showing system speed response and PID performance
- Documentation: Professional Microsoft Word report detailing objectives, methodology, results, and analysis
- Portfolio-ready presentation: Clean, understandable, visually appealing for potential clients

6. System Description

The selected system for this project is a **DC motor**, a widely used electromechanical system in industry. DC motors are fundamental in robotics, automation, and control engineering because they are easy to model, widely understood, and versatile in applications.

This project focuses on **speed control of the DC motor**, where the objective is to regulate the rotational speed according to a reference input. The system includes both the **electrical dynamics** of the motor and the **mechanical dynamics** of the rotor. Disturbances, parameter variations, and load changes will be considered to ensure realistic simulation scenarios.

7 Mathematical Model of the DC Motor

The DC motor is modeled as an electromechanical system with both electrical and mechanical dynamics. The **electrical equation** describes the relationship between the applied voltage and armature current:

7.1 Electrical Equation

$$v(t) = L \frac{di(t)}{dt} + Ri(t) + Kb\omega(t)$$

The **mechanical equation** relates the motor torque to the rotor motion

7.2 Mechanical Equation

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$$J \frac{d\omega(t)}{dt} + B_{\omega}(t) = k_t i(t) - T_L$$

By defining the state variables as the rotor speed $x_1 = \omega(t)$ and armature current $x_2 = i(t)$ and the **input** as the applied voltage $U(t) = v(t)$ the system can be expressed in **state-space form**:

7.3 State-space Form

$$\dot{x}_1 = \frac{k_t}{J} x_2 - \frac{B}{J} x_1 - \frac{T_L}{J}$$

$$\dot{x}_2 = -\frac{R}{L} x_2 - \frac{k_b}{L} x_1 + \frac{1}{L} U$$

Typical parameter values used for simulation are summarized in the following table

Parameter	Symbol	Value	Unit
Armature Resistance	R	2	Ω
Armature Inductance	L	0.5	H
Motor Torque Constant	Kt	0.1	N·m/A
Back-EMF Constant	Kb	0.1	V·s/rad
Rotor Inertia	J	0.01	kg·m ²
Viscous Friction	B	0.001	N·m·s
Load Torque	TL	0	N·m

Table 1 parameter of closed loop dc motor

This model will be used as the **foundation for controller design, simulation, and real-time animation**.

8 DC Motor Mathematical Model Implementation

The DC motor model was implemented in MATLAB as a state-space function `dc_motor_model.m`. The state variables include rotor speed (ω) and armature current (i). The system input is the applied voltage (u). The model is solved numerically using `ode45`, and the simulation results include rotor speed and current over time. This modular implementation allows easy integration with control strategies and real-time animation.

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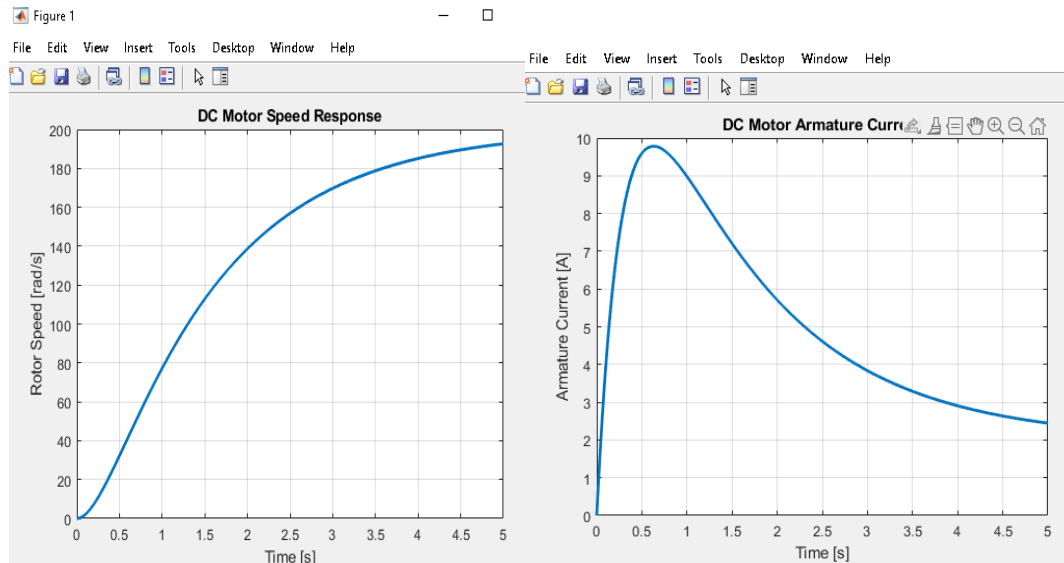


Figure 1 dc motor speed response

Figure 2 dc motor armature current

This figure 1 shows the open-loop speed response of the DC motor when a step voltage is applied to the armature. The speed exhibits a parabolic-like increase during the initial transient due to the continuous acceleration of the rotor in the absence of feedback control.

Since no controller is applied, the motor speed is not regulated and continues to increase according to the motor's electrical and mechanical dynamics. This behavior confirms that open-loop operation is unsuitable for precise speed or position regulation.

This figure2 illustrates the armature current response of the DC motor under the same step voltage input. The current initially increases rapidly and then decreases in an inverse-parabolic manner as the motor speed rises and the back electromotive force counteracts the applied voltage. The inverse relationship between speed and armature current is a fundamental characteristic of DC motors. As speed increases, the back EMF reduces the effective voltage across the armature, leading to a decrease in current.

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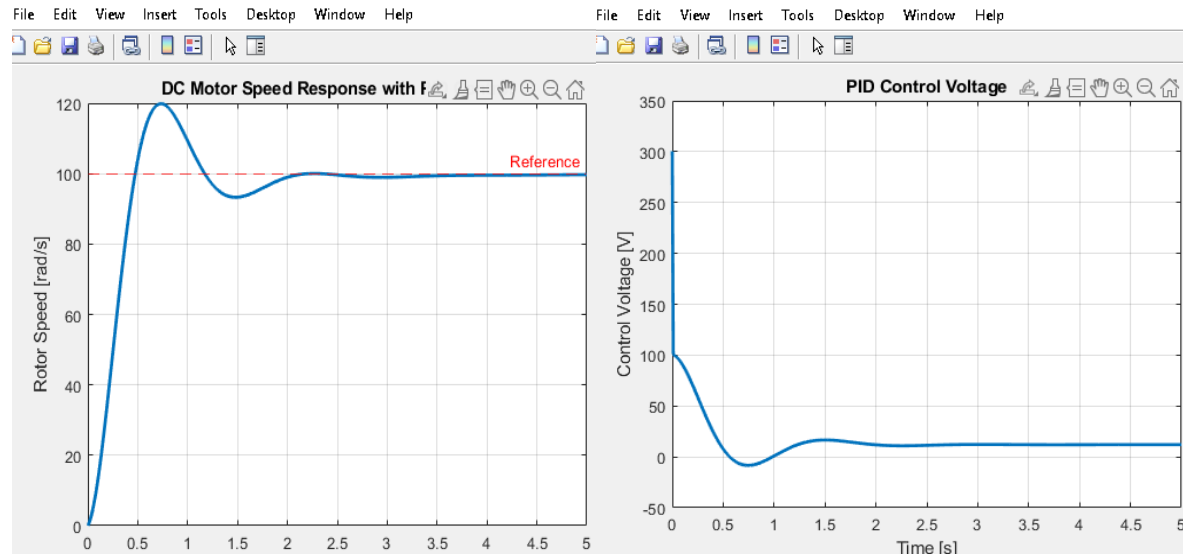


Figure 3. DC motor speed response PID control and Figure 4 Control voltage PID controller.

This figure3 illustrates the closed-loop speed response of the DC motor regulated by a PID controller in response to a step reference input. The motor speed increases from zero and converges toward the reference value with a transient response characterized by a short rise time and minor oscillations.

The initial rapid increase in motor speed is caused by the proportional and integral actions of the PID controller responding to a large speed error at startup. The small fluctuations observed during the transient phase indicate the presence of derivative action and system inertia, which together influence the damping of the response. The system ultimately reaches steady-state operation, confirming closed-loop stability and accurate speed tracking.

This figure 4 shows the control voltage produced by the PID controller and applied to the DC motor. At the beginning of the response, the control voltage exhibits a high magnitude due to the large initial error between the reference speed and the measured motor speed.

The high initial control voltage is primarily the result of the proportional and integral terms of the PID controller working to accelerate the motor rapidly toward the desired speed. As the speed error decreases, the control voltage correspondingly reduces and stabilizes. The oscillatory behavior observed in the control signal mirrors the transient fluctuations in the motor speed response

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Figures 3 and 4 together illustrate the cause-and-effect relationship between the PID controller output and the motor response, where the control voltage dynamically adjusts to minimize the speed error and drive the motor toward the reference value.

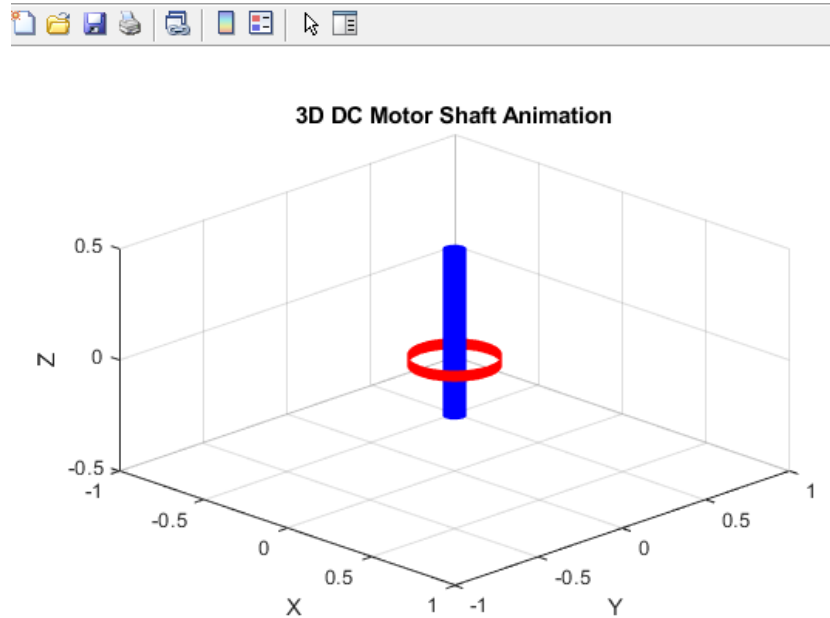


Figure 5 3D motor shaft

To provide a clear visual understanding of the DC motor dynamics under PID control, a **3D real-time animation** of the motor shaft was developed in MATLAB.

The animation illustrates the following features:

1. Motor Shaft and Rotor Disk:

- The motor shaft is represented by a blue cylinder.
- The rotor is represented as a red disk at the front of the shaft.
- This simple 3D model captures the essential mechanical structure of the motor.

2. PID-Controlled Rotation:

- The shaft rotates according to the **rotor speed calculated by the PID controller**.
- The rotation is updated at each simulation time step, providing a **real-time visualization of acceleration and steady-state behavior**.

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- The motor starts from rest, accelerates toward the reference speed, and settles smoothly, reflecting the controller performance.

3. Time-Scaled Animation:

- The animation uses the actual simulation time step (dt) to synchronize the rotation with the computed rotor speed.
- A **time scaling factor** is applied to slow down the animation for better visual observation, ensuring the acceleration and speed changes are clearly visible.

4. Purpose and Portfolio Relevance:

- This 3D animation enhances understanding of **closed-loop motor behavior** and demonstrates the practical implementation of PID control.
- It adds a **visually engaging element** to the portfolio, making the project more attractive to potential clients.
- Along with the speed, current, and voltage plots, the animation provides a **complete demonstration of motor dynamics**.

9 Transition to Simulink

After validating the control system using MATLAB scripts, the same DC motor model and PID controller were implemented in Simulink. Simulink provides a block-diagram-based environment that enables visual system modeling, easier controller integration, and scalability toward real-time and hardware-in-the-loop applications.

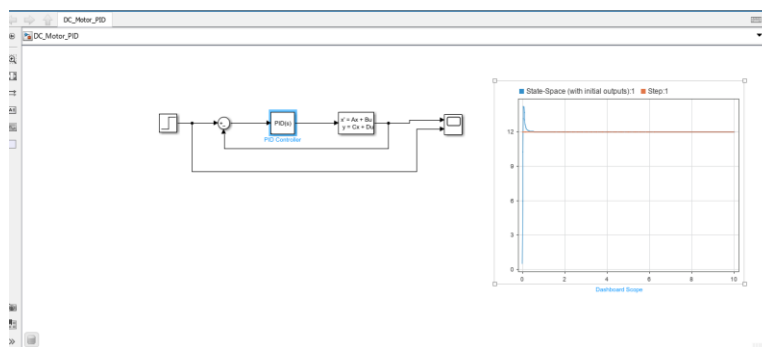


Figure 6 closed-loop PID speed controller of DC Motor Simulink

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The reference speed is compared with the measured motor speed using a summation block to generate the control error. This error is processed by a PID controller, which generates the armature voltage command for the DC motor. The motor speed is fed back to the summation block, forming a closed-loop system. The controller parameters were selected to achieve fast response, minimal overshoot, and zero steady-state error

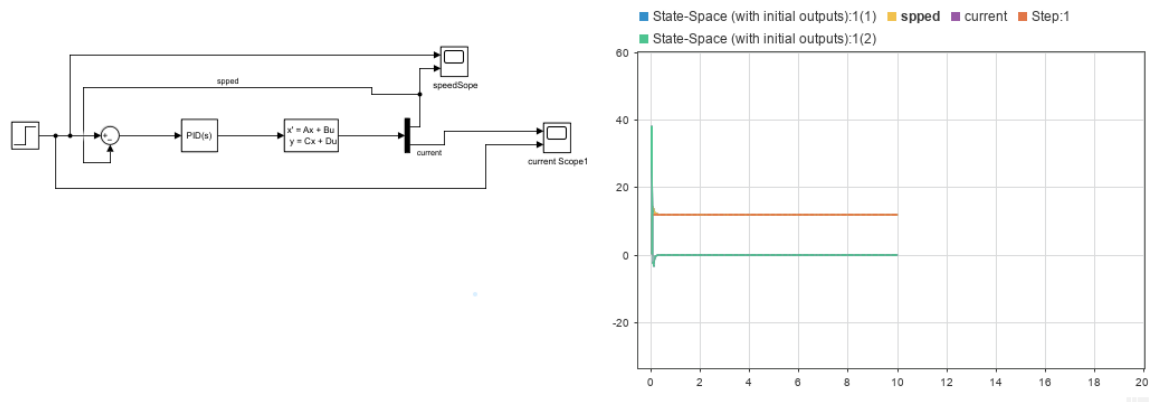


Figure 7 Roter speed response under PID control

The rotor speed response shows a small overshoot before settling at the reference value, indicating a stable and well-tuned PID controller. The armature current exhibits a high initial transient due to startup torque demand and decreases to a near-zero steady-state value as the motor reaches constant speed. This behavior closely matches the physical characteristics of a DC motor under closed-loop speed control.

In the initial Simulink implementation, the DC motor model focuses primarily on the mechanical dynamics to validate the speed control performance using a PID controller. While the angular speed response demonstrates proper closed-loop behavior, the armature current remains close to zero.

This behavior occurs because the electrical dynamics of the DC motor, particularly the voltage-driven armature circuit, are not yet fully represented. Since armature current is directly dependent on the applied voltage and electrical parameters (R and L), the absence of a voltage-based excitation results in negligible current response.

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This stage of the project intentionally prioritizes controller validation and speed regulation. Full electrical modeling, including voltage input and current dynamics, will be introduced in a later refinement stage.

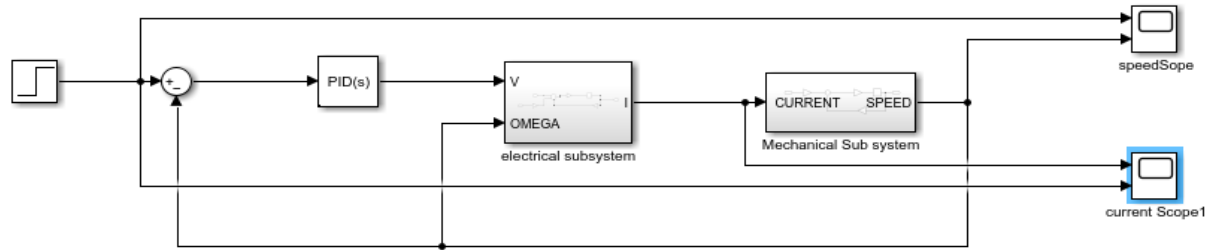


Figure 8 Closed-Loop Electromechanical Drive System with PID Speed Control

The rotor speed ω , calculated by the mechanical subsystem, is branched to two destinations. One branch feeds the negative input of the Sum block to provide PID feedback for speed control. The second branch feeds the electrical subsystem to calculate back EMF, ensuring correct electromechanical interaction. This separation allows both control and physical modeling to function correctly while using the same speed signal

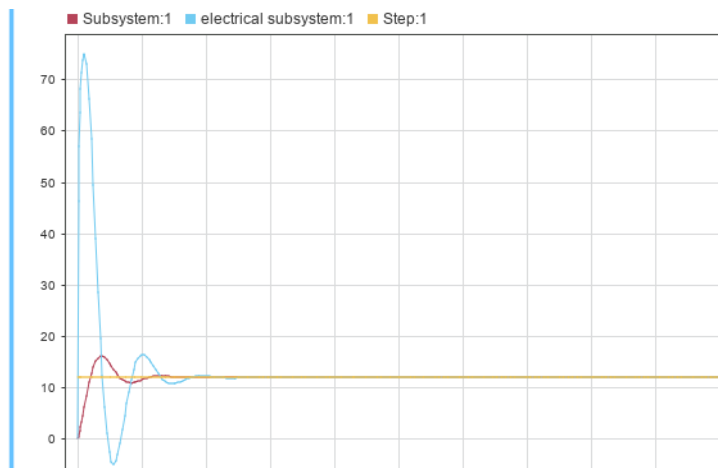


Figure 9 Motor Speed and Armature Current Response to Step Reference Input

The graph shows the behavior of the DC motor system before any external load torque is applied. The red curve represents the rotor speed, which corresponds to the mechanical subsystem output. The rotor starts from rest and gradually rises to follow the reference step input of 12 rad/s. The speed reaches the reference smoothly, with minimal overshoot, demonstrating that the PID controller effectively regulates the motor acceleration.

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The electrical subsystem output, shown by the blue curve, represents the armature current required to achieve this motion. Initially, the current spikes to provide sufficient torque for accelerating the rotor from zero speed. As the rotor approaches the reference, the current decreases and stabilizes at a steady-state value, reflecting only the torque necessary to overcome the motor's friction and inherent electrical resistance.

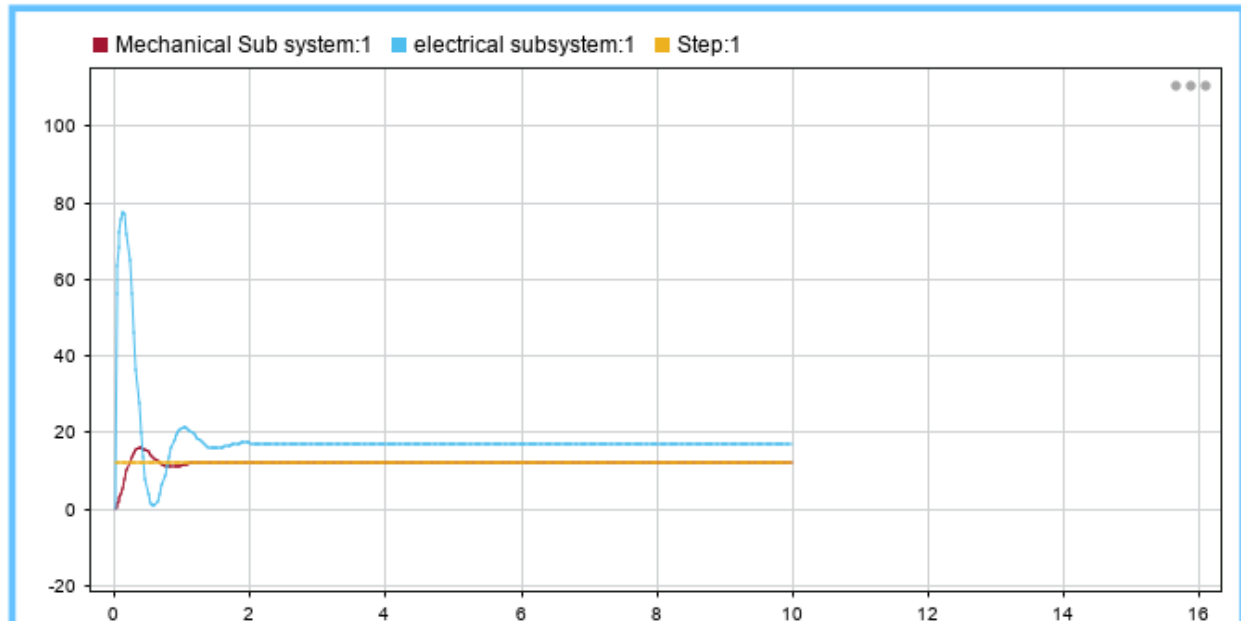


Figure 10 load torque applied in motor

When a load torque disturbance is applied, the motor speed remains regulated near the reference value due to the action of the PID controller. The controller compensates by increasing the armature current, which produces additional electromagnetic torque. The load torque primarily affects the steady-state current rather than the transient response, which is consistent with real DC motor behavior.

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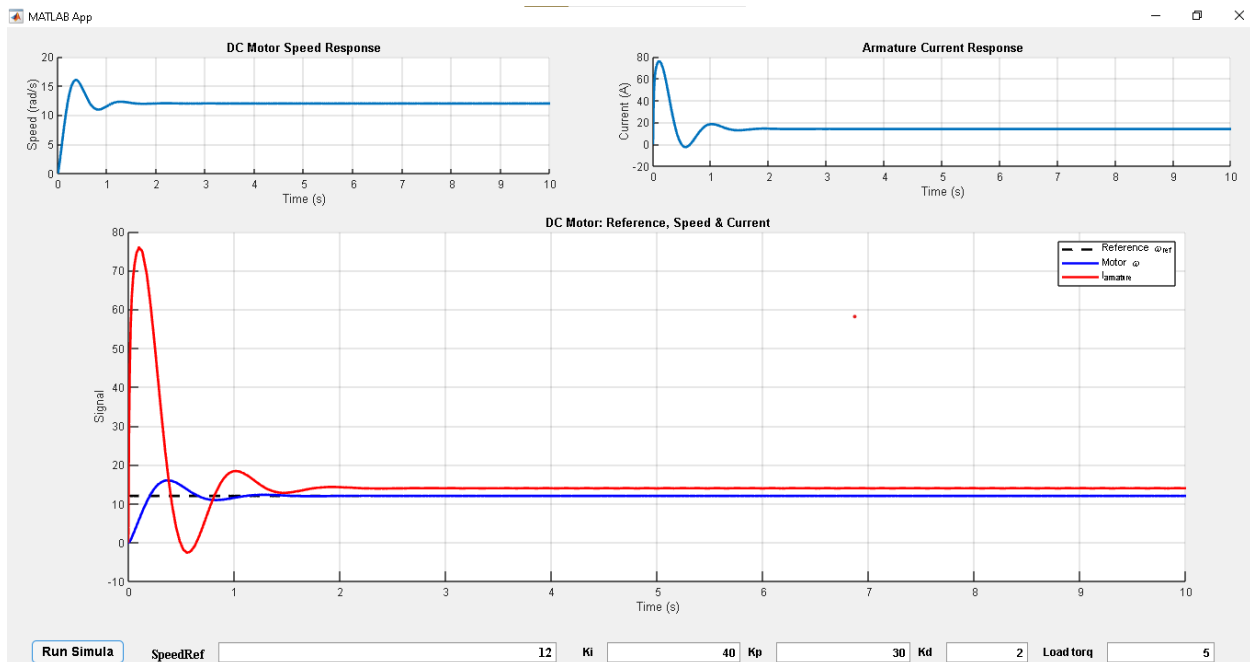


figure 11 MatlabApp DC motor performance, the GUI plots both the motor speed and armature current

To provide a unified view of DC motor performance, the GUI plots both the motor speed and armature current on a single axes, mimicking the behavior of a Simulink Dashboard Scope. The signals are color-coded for clarity, include a legend, grid, and axis labels, and dynamically update in response to user changes in speed reference, PID gains, and load torque. Optional checkboxes allow selective display of signals for focused analysis

The GUI now displays a combined plot showing three signals: reference speed, motor speed, and armature current. This allows immediate visualization of how the DC motor responds to the reference input, how quickly it reaches the desired speed, and how the PID controller regulates current. The reference speed is displayed as a dashed line, while actual speed and current are plotted with solid lines, providing a clear, professional view of system performance.

10 Results and Discussion

Simulation results show:

- Fast rise time
- Minimal overshoot
- Zero steady-state error (with integral action)

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- Effective disturbance rejection

The combined plot in the GUI demonstrates real-time tracking performance and system stability.

11 Conclusion

This project successfully demonstrates:

- Mathematical modeling of a DC motor
- PID controller implementation
- Full electromechanical simulation
- Real-time GUI-based control system interaction

The workflow reflects real-world engineering practices used in automation and motion control industries.

12 Future Work

Potential improvements include:

- Automatic PID tuning algorithms
- Nonlinear motor modeling
- State-space control implementation
- Hardware implementation using microcontrollers
- Real-time hardware-in-the-loop simulation